Ningzhe Shi

Ph.D. Candidate in Computer Science and Technology

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Research Profile

Ph.D. candidate researching 6G intelligent networks and AI service orchestration, with a focus on communication, sensing, computing, memory, and knowledge-resource optimization through reinforcement learning and mathematical optimization.

Education

Ph.D. Candidate **Institute of Computing Technology, Chinese Academy of Sciences; University of Chinese Academy of Sciences**, Computer Science and Technology (A+ national discipline) — Beijing, China; Sept 2021 – present
- Ph.D. Advisor: Prof. Yiqing Zhou

B.Eng. **Xidian University**, Communication Engineering (A+ national discipline) — Xi'an, China; Sept 2017 – July 2021
- Undergraduate Advisor: Prof. Junyu Liu
- Ranked 1st in class
- National Scholarship recipient

Internship Experience

China Mobile, Digital Intelligence Department, AI Agent Algorithm Intern — China; Jan 2026 – Mar 2026
- Investigated long-horizon memory mechanisms for AI agents in multi-turn interactions, focusing on context forgetting, historical redundancy, and response consistency.
- Surveyed retrieval-augmented memory, hierarchical storage, self-evolving memory updates, memory extraction, compression, forgetting, and dynamic updating.
- Reproduced and analyzed the Mem-alpha memory architecture and formulated memory selection, compression, and updating as a policy-optimization problem.
- Studied reinforcement-learning-based memory maintenance and memory-aware LLM task scheduling under end-edge-cloud collaboration.

Research Experience

Institute of Computing Technology, Chinese Academy of Sciences, Main Contributor, Knowledge-Matching-Aware Scheduling for RAG Services — Oct 2025 – June 2026
- Modeled task semantics, node knowledge coverage, computing cost, transmission delay, generation quality, and energy consumption for cloud-edge RAG services.
- Designed a knowledge-matching-aware RAG task scheduling mechanism under heterogeneous knowledge bases and dynamic resource states.
- Built a simulation environment; reduced energy consumption by approximately 50% and service delay by more than 10% in evaluated settings.

Institute of Computing Technology, Chinese Academy of Sciences, Main Contributor, Memory-Aware Distributed Routing for Space AI Computing Networks — Oct 2025 – Mar 2026
- Modeled the coupling among task routing, node memory state, onboard computing load, storage availability, and link conditions.
- Designed a distributed routing method for high-dimensional dynamic network states and introduced diffusion-assisted policy generation.
- Validated the method on a ground-based space computing network simulator, reducing end-to-end routing delay by more than 15%.

Institute of Computing Technology, Chinese Academy of Sciences, Main Contributor, Distributed Quantum Reinforcement Learning for AIGC Cloud-Edge Scheduling — June 2025 – Dec 2025

- Modeled collaborative cloud-edge processing for multi-user AIGC services, coupling task offloading, computing allocation, and service delay.
- Designed a distributed hybrid quantum-classical reinforcement learning scheduler using parameterized quantum circuits and classical neural networks.
- Reduced model parameters by more than 95% and average service delay by more than 30% compared with baseline methods in simulation.

Institute of Computing Technology, Chinese Academy of Sciences, Main Contributor and First Author, ISAC-Driven AIGC Service Orchestration — Jan 2024 – June 2025

- Designed a content-accuracy-and-quality-aware service metric and modeled sensing, computing, communication, and AIGC service experience as a coupled system.
- Formulated the joint resource-allocation problem and designed an LP-guided hierarchical reinforcement learning framework.
- Improved average user service experience by more than 50% in evaluated settings; related work was published in IEEE Transactions on Mobile Computing.

Institute of Computing Technology, Chinese Academy of Sciences, Main Contributor and First Author, User Selection and Resource Allocation for High-Concurrency AI Services — Dec 2022 – Dec 2024

- Modeled user access, bandwidth, subchannels, transmit power, edge computing resources, and service satisfaction in multi-cell MEC networks.
- Designed a joint optimization algorithm for user selection, channel assignment, power control, and edge computing resource allocation.
- Improved average user satisfaction by more than 50%; related work was published in IEEE Transactions on Mobile Computing.

Institute of Computing Technology, Chinese Academy of Sciences, Main Contributor and First Author, Adaptive Fair Resource Scheduling Under Imperfect CSI — May 2022 – May 2024

- Built a channel-uncertainty-aware resource allocation model for multi-cell multi-user downlink communications under imperfect CSI.
- Designed a reward-adaptive multi-agent DRL scheduler balancing long-term fairness and network service rate.
- Improved long-term fairness by 13% and service rate by more than 10%; related work was published in IEEE Transactions on Cognitive Communications and Networking.

Institute of Computing Technology, Chinese Academy of Sciences, Technical Lead, Multi-Channel Access for In-Band Full-Duplex Ad Hoc Networks — Oct 2022 – June 2023

- Designed multi-channel interference coordination and access-control mechanisms for full-duplex ad hoc networks with residual self-interference and inter-node interference.
- Built a distributed WebSocket-based system-level simulator integrating node workers, channel computation, mobility modeling, and visualization.
- Delivered a system prototype and related invention patent/software copyright outputs.

Institute of Computing Technology, Chinese Academy of Sciences, Task Leader, Technical Lead, and First Author, UAV-Assisted Secure Transmission Under Deception and Jamming — Sept 2021 – June 2023

- Built a UAV-assisted mmWave MIMO secure communication model covering legitimate links, deception links, jamming sources, and UAV-assisted nodes.
- Designed joint optimization for beamforming, UAV three-dimensional placement, and power allocation under complex electromagnetic confrontation.
- Improved secure transmission rate to approximately two times that of the no-deception baseline; related work was published in China Communications and one invention patent was granted.

Publications

Service Satisfaction Based User Selection and Resource Allocation for NOMA-Based Multi-Cell MEC Networks 2026
Ningzhe Shi, Yiqing Zhou, Ling Liu, Yihao Wu, Hanxiao Yu, Jinglin Shi
10.1109/TMC.2026.3673357 (IEEE Transactions on Mobile Computing)

Content Accuracy and Quality Aware Resource Allocation Based on LP-Guided DRL for ISAC-Driven AIGC Networks 2026
Ningzhe Shi, Yiqing Zhou, Ling Liu, Jinglin Shi, Yihao Wu, Haiwei Shi, Hanxiao Yu
10.1109/TMC.2026.3652144 (IEEE Transactions on Mobile Computing)

Long-Term Fairness From Real-Time Decisions: Reward Adaptive Multi-Agent DRL for Wireless Communication With Imperfect CSI 2026
Ningzhe Shi, Ling Liu, Yiqing Zhou, Hanxiao Yu, Yihao Wu, Jingya Yang, Jinglin Shi
10.1109/TCCN.2026.3692166 (IEEE Transactions on Cognitive Communications and Networking)

Deception-Based Secure Communication for UAV-Assisted mmWave MIMO Systems 2026
Ningzhe Shi, Yiqing Zhou, Yu Zhang, Zhijun Han, Ling Liu, Jinglin Shi
10.23919/JCC.fa.2025-0337.202604 (China Communications)

Cross-Enabling of Operation, Information and Communication: Building the Foundation for Intelligent New-Generation Productivity 2025
Ningzhe Shi, Yihao Wu, Yaxing Xu, Qing Cai, Yiqing Zhou
10.11991/cccf.202508013 (Computing Magazine of the CCF)

Patents & Software Copyright

Construction Method and Resource Management Method for a Wireless Network Resource Allocation System Apr 2026
Granted invention patent: CN202310354794.0

Transmission Device and Semantic Transmission Method for High-Mobility Communications Mar 2026
Invention patent application: CN202610406457.5

Joint Channel Estimation and Signal Detection Method, Signal Receiver, and Storage Medium Oct 2025
Invention patent application: CN202511536493.5

Resource Allocation Method for AI-Generated Content Services Based on Integrated Sensing and Communication July 2025
Invention patent application: CN202511053332.0

Environment- and Resource-State-Aware Resource Allocation Method and System July 2025
Invention patent application: CN202510955830.8

Mobile Ad Hoc Network Data Routing Method and Industrial Ad Hoc Network System June 2025
Invention patent application: CN202510873622.3

Anti-Jamming Method and System Based on Relay Position Selection and Power Allocation May 2025
Granted invention patent: CN202310078808.0

Anti-Tracking Jamming-Resistant Signal Transmission Method, Signal Reception Method, and System Sept 2024
Granted invention patent: CN202211134854.X

Channel Access Method and Communication System for Ad Hoc Networks Mar 2024
Invention patent application: CN202410348337.5

Multi-User MIMO Signal Detection Model Jan 2024
Invention patent application: CN202410018551.4

Large-Scale Full-Duplex Network Simulation System Jan 2024
Registered software copyright: 2024SR0082117

Real-Time Simulation System and Method for Full-Duplex Ad Hoc Networks Dec 2023
Invention patent application: CN202311719290.0

Construction Method and Resource Management Method for a Wireless Network Resource Allocation System Apr 2023
Invention patent application (PCT): PCT/CN2023/090283

Honors and Awards

Outstanding Graduate Student, Xinghang Program June 2026
Journal of Electronics & Information Technology; national selection of 22 graduate students

Outstanding Poster Award June 2026
Frontiers in Electronics and Information Conference

UCAS Outstanding Student Model June 2026
University-wide top 1%

ICT Outstanding Student Jan 2026
Institute of Computing Technology / State Key Laboratory of Processors

E Fund Financial Technology Doctoral Scholarship Jan 2025
12 recipients institute-wide

UCAS Outstanding Student Cadre May 2025

Outstanding Volunteer, Chinese Academy of Sciences Public Science Day 2023-2025

UCAS Outstanding Student June 2023

UCAS First-Class Scholarship 2021-2025
Awarded for five consecutive years

Outstanding Graduate, Xidian University June 2021

First-Class Graduation Scholarship, Xidian University June 2021

Honorable Mention, Mathematical Contest in Modeling 2020

National Scholarship 2019

National Second Prize, China Undergraduate Mathematical Contest in Modeling 2019

National Encouragement Scholarship 2018

Third Prize, CETC-10 Communication Cup 2019

Academic Service

Program Committee: Technical Program Committee Member, IEEE International Conference on Communications (ICC), 2026

Academic Correspondence: Academic Correspondent, Digital Communications and Networks

Peer Review: IEEE Transactions on Mobile Computing (TMC); IEEE Transactions on Communications (TCOM); IEEE Internet of Things Journal; IEEE Transactions on Vehicular Technology (TVT); Science China Information Sciences; China Communications; IEEE International Conference on Communications (ICC); IEEE Vehicular Technology Conference (VTC); Journal on Communications; Journal of Electronics & Information Technology

Outreach and Public Engagement

Science Outreach

- Science outreach volunteer, Chinese Academy of Sciences Public Science Day
- Committee Member, UCAS Science Popularization Association
- Participated in public science communication, research demonstrations, and science outreach activities
- Outstanding Science Popularization Volunteer, Bureau of Academic Divisions, Chinese Academy of Sciences

Leadership and Student Service

Graduate Stage

- Vice President, Haidian Branch of the Volunteer Federation
- Committee Member, UCAS Science Popularization Association
- Class Mental Health Liaison

Undergraduate Stage

- Youth League Branch Secretary
- Team Leader, Horizon Volunteer Team, School of Telecommunications Engineering, Xidian University

Skills

AI-Native Network Research: 6G intelligent networks; cloud-edge collaborative AI services; RAG service orchestration; AI Agent memory systems; space AI computing networks; ISAC-driven AIGC services

Optimization and Learning: Reinforcement learning; multi-agent DRL; LP-guided hierarchical scheduling; diffusion-assisted policy generation; quantum-classical reinforcement learning; convex optimization; robust and fairness-aware resource allocation

Wireless and Computing Systems: MEC/NOMA networks; full-duplex ad hoc networks; UAV-assisted secure communications; integrated sensing, communication, computing, and knowledge-resource optimization

Implementation and Simulation: Python; PyTorch; MATLAB; C++; Shell; custom RL environments; system-level wireless simulation; distributed WebSocket-based simulation; experiment automation and visualization

Research Delivery: Problem formulation; algorithm design; simulation validation; ablation studies; technical reporting; journal paper writing; patent and software-copyright preparation; cross-team technical coordination